AILIN POLLIO S. P.

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PROFILE

ML Research Engineer working at the intersection of deep learning, speech, and developmental AI. Experienced in developing models that capture meaningful vocal and linguistic patterns.

EXPERIENCE

Research Engineer
Centre National de la Recherche Scientifique (CNRS) Apr 2026 - Sep 2026

· End-to-end dysarthric speech reconstruction

AI Research Engineer
Virtuosis AI (EPFL spin-off) Jun 2025 - Dec 2025

· Developed and evaluated AI models for vocal biomarker detection across neurological and mental health conditions (Parkinson's, Alzheimer's, MCI, stress, depression).
· Built multilingual speech pipelines using handcrafted acoustic features (openSMILE, Praat) and self-supervised embeddings (HuBERT, Wav2Vec2, BYOL-S, Google HEAR).
· Contributed to scientific writing, data analysis, and validation workflows with academic and clinical collaborators.
· Worked with Microsoft Azure services, including Blob Storage, Cosmos DB (Mongo API), Azure Machine Learning, and cloud-based pipelines for large-scale audio processing and model deployment.

Machine Learning Engineer Intern
Computational Speech Group (University of Eastern Finland) Jan 2025 – Jun 2025

· Developed models to predict articulatory movements from speech using Wav2Vec2.0 and EMA data.
· Conducted probing and analysis under the supervision of Prof. Tomi Kinnunen and Dr. Alexandre Nikolaev.

NLP Engineer Intern (Speech-to-Text)
Callyope May 2024 – Aug 2024

· Built pipelines for speech-to-text alignment of clinical recordings (depression vs. control).
· Generated Praat TextGrid annotations for linguistic and acoustic analysis.

Linguistic Data Specialist
Appen / Home Project.Tech 2020 - 2023

· Produced and reviewed high-accuracy phonemic transcriptions (IPA, SAMPA).
· Annotated domain-specific speech and text data to improve AI assistant performance.
· Supported post-editing and quality control for multilingual transcription projects.

EDUCATION

PhD in Computer Science 2023-2025
Aix-Marseille University
Laboratoire d'Informatique et Systèmes (LIS), TALEP group, France

European Master's in Clinical Linguistics (EMCL++) 2023-2025
University of Groningen, The Netherlands
University of Eastern Finland, Finland
Ghent University, Belgium
Thesis grade: A

Diploma in Python 3 2024
National Technological University, Argentina
Final grade: Excellent

Specialization in Cognitive Neuroscience of Language and Reading 2022-2023
University of Buenos Aires, Argentina

Bachelor in Linguistics and Literature 2021
Orientation in Formal Linguistics
University of Mar del Plata, Argentina.
CGPA: 8.9/10

PUBLICATIONS

Pollio San Pedro, A., Kinnunen, T., Nikolaev, A., Pandey, R. *How Bilingual Are SSL Speech Models? Cross-Lingual Probing of Articulatory Encoding with Finnish and Russian EMA.*
Proc. Interspeech 2026, 3553-3557, doi: 10.21437/Interspeech.2026-1324 2026

Pollio San Pedro, A. *The Presence of Articulatory Representations in Self-Supervised Speech Models: A Cross-Linguistic Perspective*
DOI: 10.13140/RG.2.2.28626.41926
Thesis for: Master's in Clinical Linguistics
Advisors: Prof. Dr. Tomi Kinnunen and Dr. Alexandre Nikolaev 2025

SCHOLARSHIPS AWARDED

- EMJMD for the European Master's in Clinical Linguistics (EMCL++) 2023 - 2025
- Research Scholarship, Postgraduate Type "A", National University of Mar del Plata
  Project: "Relationships Between Syntactic Skills, Reading Skills, and Working Memory" 2023
- Research Scholarship, "Advanced Student" Category, National University of Mar del Plata 2021
- Research Scholarship, "Advanced Student" Category, National University of Mar del Plata 2020
- EVC-CIN Research Scholarship, Ministry of Education, Argentina 2019

LANGUAGES

Spanish	Native
English	Full professional proficiency
Italian	Full professional proficiency
German	Basic knowledge
French	Basic knowledge

KEY WORDS

Signal Processing - Python - Pandas - Scikit-learn - Azure - Hugging Face - Speech Processing - Machine Learning - Deep Learning - Self-Supervised Learning - Vocal Biomarkers - openSMILE - Praat - Wav2Vec2 - HuBERT - BYOL- S - PyTorch - Feature Extraction - Cross-Lingual Modeling - Research Writing - Scientific Communication